

JEE Mains 2019 Chapter wise Question Bank

Point and Straight Line – Questions

Q1

Consider the set of all lines $px + qy + r = 0$ such that $3p + 2q + 4r = 0$. Which one of the following statements is true?

- (1) The lines are concurrent at the point $\left(\frac{3}{4}, \frac{1}{2}\right)$.
- (2) Each line passes through the origin.
- (3) The lines are all parallel.
- (4) The lines are not concurrent.

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Q2

Let S be the set of all triangles in the xy -plane, each having one vertex at the origin and the other two vertices lie on coordinate axes with integral coordinates. If each triangle in S has area 50 sq. units, then the number of elements in the set S is:

- (1) 9 (2) 18
(3) 36 (4) 32

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Q3

Let the equations of two sides of a triangle be $3x - 2y + 6 = 0$ and $4x + 5y - 20 = 0$. If the orthocentre of this triangle is at $(1, 1)$, then the equation of its third side is:

- (1) $122y - 26x - 1675 = 0$
(2) $122y + 26x + 1675 = 0$
(3) $26x + 61y + 1675 = 0$
(4) $26x - 122y - 1675 = 0$

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Q4

If the lines $x = ay + b$, $z = cy + d$ and $x = a'z + b'$, $y = c'z + d'$ are perpendicular, then:

- (1) $ab' + bc' + 1 = 0$
(2) $cc' + a + a' = 0$
(3) $bb' + cc' + 1 = 0$
(4) $aa' + c + c' = 0$

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Q5

A point P moves on the line $2x - 3y + 4 = 0$. If $Q(1, 4)$ and $R(3, -2)$ are fixed points, then the locus of the centroid of ΔPQR is a line:

- (1) with slope $\frac{3}{2}$
(2) parallel to x -axis
(3) with slope $\frac{2}{3}$
(4) parallel to y -axis

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Q6

If the line $3x + 4y - 24 = 0$ intersects the x -axis at the point A and the y -axis at the point B , then the incentre of the triangle OAB , where O is the origin, is:

- (1) $(3, 4)$ (2) $(2, 2)$
(3) $(4, 3)$ (4) $(4, 4)$

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Q7

Two sides of a parallelogram are along the lines, $x + y = 3$ and $x - y + 3 = 0$. If its diagonals intersect at $(2, 4)$, then one of its vertex is:

- (1) $(3, 5)$ (2) $(2, 1)$
(3) $(2, 6)$ (4) $(3, 6)$

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Point and Straight Line

Q8

Two vertices of a triangle are (0, 2) and (4, 3). If its orthocentre is at the origin, then its third vertex lies in which quadrant?

- (1) third (2) second
(3) first (4) fourth

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Q9

The straight line $x + 2y = 1$ meets the coordinate axes at A and B. A circle is drawn through A, B and the origin. Then the sum of perpendicular distances from A and B on the tangent to the circle at the origin is :

- (1) $\frac{\sqrt{5}}{2}$ (2) $2\sqrt{5}$
(3) $\frac{\sqrt{5}}{4}$ (4) $4\sqrt{5}$

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Q10

If in a parallelogram ABDC, the coordinates of A, B and C are respectively (1, 2), (3, 4) and (2, 5), then the equation of the diagonal AD is :

- (1) $5x - 3y + 1 = 0$ (2) $5x + 3y - 11 = 0$
(3) $3x - 5y + 7 = 0$ (4) $3x + 5y - 13 = 0$

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Q11

A tetrahedron has vertices P(1, 2, 1), Q(2, 1, 3), R(-1, 1, 2) and O(0, 0, 0). The angle between the faces OPQ and PQR is:

- (1) $\cos^{-1}\left(\frac{17}{31}\right)$ (2) $\cos^{-1}\left(\frac{19}{35}\right)$
(3) $\cos^{-1}\left(\frac{9}{35}\right)$ (4) $\cos^{-1}\left(\frac{7}{31}\right)$

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Q12

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If the straight line, $2x - 3y + 17 = 0$ is perpendicular to the line passing through the points (7, 17) and (15, β), then β equals :

- (1) $\frac{35}{3}$ (2) -5
(3) $-\frac{35}{3}$ (4) 5

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Q13

If a straight line passing through the point P(-3, 4) is such that its intercepted portion between the coordinate axes is bisected at P, then its equation is :

- (1) $3x - 4y + 25 = 0$ (2) $4x - 3y + 24 = 0$
(3) $x - y + 7 = 0$ (4) $4x + 3y = 0$

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Q14

A point on the straight line, $3x + 5y = 15$ which is equidistant from the coordinate axes will lie only in :

- (1) 4th quadrant
(2) 1st quadrant
(3) 1st and 2nd quadrants
(4) 1st, 2nd and 4th quadrants

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Q15

Let O(0, 0) and A(0, 1) be two fixed points. Then the locus of a point P such that the perimeter of ΔAOP is 4, is :

- (1) $8x^2 - 9y^2 + 9y = 18$
(2) $9x^2 - 8y^2 + 8y = 16$
(3) $9x^2 + 8y^2 - 8y = 16$
(4) $8x^2 + 9y^2 - 9y = 18$

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Q16

Suppose that the points (h, k), (1, 2) and (-3, 4) lie on the line L_1 . If a line L_2 passing through the points (h, k) and (4, 3) is perpendicular on L_1 , then $\frac{k}{h}$ equals :

- (1) $\frac{1}{3}$ (2) 0 (3) 3 (4) $-\frac{1}{7}$

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Point and Straight Line

Q17

Slope of a line passing through P(2, 3) and intersecting the line $x + y = 7$ at a distance of 4 units from P, is:

- (1) $\frac{1-\sqrt{5}}{1+\sqrt{5}}$ (2) $\frac{1-\sqrt{7}}{1+\sqrt{7}}$
(3) $\frac{\sqrt{7}-1}{\sqrt{7}+1}$ (4) $\frac{\sqrt{5}-1}{\sqrt{5}+1}$

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Q18

If the two lines $x + (a-1)y = 1$ and $2x + a^2y = 1$ ($a \in \mathbb{R} - \{0, 1\}$) are perpendicular, then the distance of their point of intersection from the origin is:

- (1) $\frac{\sqrt{2}}{5}$ (2) $\frac{2}{5}$ (3) $\frac{2}{\sqrt{5}}$ (4) $\frac{\sqrt{2}}{5}$

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Q19

A rectangle is inscribed in a circle with a diameter lying along the line $3y = x + 7$. If the two adjacent vertices of the rectangle are $(-8, 5)$ and $(6, 5)$, then the area of the rectangle (in sq. units) is:

- (1) 84 (2) 98 (3) 72 (4) 56

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Q20

Lines are drawn parallel to the line $4x - 3y + 2 = 0$, at a distance $\frac{3}{5}$ from the origin. Then which one of the following points lies on any of these lines?

- (1) $\left(-\frac{1}{4}, \frac{2}{3}\right)$ (2) $\left(\frac{1}{4}, -\frac{1}{3}\right)$
(3) $\left(\frac{1}{4}, \frac{1}{3}\right)$ (4) $\left(-\frac{1}{4}, -\frac{2}{3}\right)$

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Q21

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The equation $y = \sin x \sin(x+2) - \sin^2(x+1)$ represents a straight line lying in :

- (1) second and third quadrants only
(2) first, second and fourth quadrant
(3) first, third and fourth quadrants
(4) third and fourth quadrants only

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Q22

A triangle has a vertex at (1, 2) and the mid points of the two sides through it are $(-1, 1)$ and $(2, 3)$. Then the centroid of this triangle is :

- (1) $\left(1, \frac{7}{3}\right)$ (2) $\left(\frac{1}{3}, 2\right)$ (3) $\left(\frac{1}{3}, 1\right)$ (4) $\left(\frac{1}{3}, \frac{5}{3}\right)$

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Q23

A straight line L at a distance of 4 units from the origin makes positive intercepts on the coordinate axes and the perpendicular from the origin to this line makes an angle of 60° with the line $x + y = 0$. Then an equation of the line L is:

- (1) $x + \sqrt{3}y = 8$
(2) $(\sqrt{3}+1)x + (\sqrt{3}-1)y = 8\sqrt{2}$
(3) $\sqrt{3}x + y = 8$
(4) $(\sqrt{3}-1)x + (\sqrt{3}+1)y = 8\sqrt{2}$

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Point and Straight Line - Solutions

Q1**(1)** The given equations of the set of all lines

$$px + qy + r = 0 \quad \dots(1)$$

and given condition is :

$$3p + 2q + 4r = 0$$

$$\Rightarrow \frac{3}{4}p + \frac{2}{4}q + r = 0 \quad \dots(2)$$

From (1) & (2) we get :

$$\therefore x = \frac{3}{4}, y = \frac{1}{2}$$

Hence the set of lines are concurrent and passing through

the fixed point $\left(\frac{3}{4}, \frac{1}{2}\right)$ **(3)** One of the possible ΔOAB is $A(a, 0)$ and $B(0, b)$.

$$\text{Area of } \Delta OAB = \frac{1}{2} |ab|.$$

$$\therefore |ab| = 100$$

$$|a| |b| = 100$$

But $100 = 1 \times 100, 2 \times 50, 4 \times 25, 5 \times 20$ or 10×10 $\therefore$ For $1 \times 100, a = 1$ or -1 and $b = 100$ or -100 $\therefore$ Total possible pairs are 8.Total possible pairs for $1 \times 100, 2 \times 50, 4 \times 25$ or 5×20 are 4×8 .And for 10×10 total possible pairs are 4. $\therefore$ Total number of possible triangles with integral coordinates are $4 \times 8 + 4 = 36$.

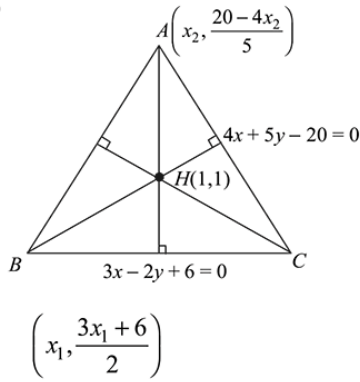
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Q2**Q3**

Point and Straight Line

(4)



Since, AH is perpendicular to BC

Hence, $m_{AH} \cdot m_{BC} = -1$

$$\left(\frac{\frac{20-4x_2}{5} - 1}{x_2 - 1} \right) \times \frac{3}{2} = -1$$

$$\frac{15-4x_2}{5(x_2-1)} = -\frac{2}{3}$$

$$45 - 12x_2 = -10x_2 + 10$$

$$2x_2 = 35 \Rightarrow x_2 = \frac{35}{2}$$

$$\Rightarrow A\left(\frac{35}{2}, -10\right)$$

Since, BH is perpendicular to CA.

Hence, $m_{BH} \times m_{CA} = -1$

$$\left(\frac{\frac{3x_1}{2} + 3 - 1}{x_1 - 1} \right) \left(-\frac{4}{5} \right) = -1$$

$$\frac{(3x_1 + 4)}{2(x_1 - 1)} \times 4 = 5$$

$$\Rightarrow 6x_1 + 8 = 5x_1 - 5 \Rightarrow x_1 = -13 \Rightarrow \left(-13, \frac{-33}{2} \right)$$

$\Rightarrow$ Equation of line AB is

$$y + 10 = \left(\frac{-\frac{33}{2} + 10}{-13 - \frac{35}{2}} \right) \left(x - \frac{35}{2} \right)$$

$$\Rightarrow -61y - 610 = -13x + \frac{455}{2}$$

$$\Rightarrow -122y - 1220 = -26x + 455$$

$$\Rightarrow 26x - 122y - 1675 = 0$$

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Q4

(4) First line is: $x = ay + b, z = cy + d$

$$\frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c}$$

and another line is: $x = a'z + b', y = c'z + d'$

$$\Rightarrow \frac{x-b'}{a'} = \frac{y-d'}{c'} = \frac{z}{1}$$

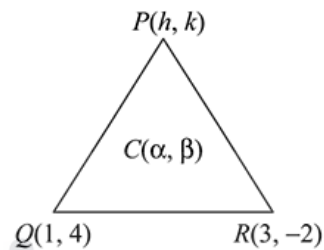
$\therefore$ Both lines are perpendicular to each other

$$\therefore aa' + c' + c = 0$$

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Q5

(3)



Let centroid C be (α, β)

$$\text{we have } \alpha = \frac{1+3+h}{3} \Rightarrow h = 3\alpha - 4$$

$$\beta = \frac{4-2+k}{3} \Rightarrow k = 3\beta - 2$$

but $P(h, k)$ lies on $2x - 3y + 4 = 0$

$$\Rightarrow 2(3\alpha - 4) - 3(3\beta - 2) + 4 = 0$$

$$\Rightarrow 6\alpha - 9\beta - 8 + 6 + 4 = 0$$

$$\Rightarrow 6\alpha - 9\beta + 2 = 0$$

$$\text{Locus: } 6x - 9y + 2 = 0$$

$$\Rightarrow y = \frac{6}{9}x + \frac{2}{9}$$

$$\therefore \text{ its slope} = \frac{6}{9} = \frac{2}{3}$$

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Q6

Point and Straight Line

(2) Equation of the line is:

$$3x + 4y = 24$$

$$\frac{x}{8} + \frac{y}{6} = 1$$

∴ coordinates of A, B & O are (8, 0), (0, 6) & (0, 0) respectively.

$$\Rightarrow OA = 8, OB = 6 \text{ \& } AB = 10.$$

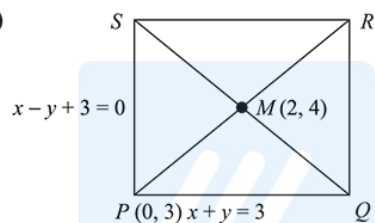
∴ Incentre of $\triangle OAB$ is given as:

$$I \equiv \left(\frac{8 \times 0 + 6 \times 8 + 10 \times 0}{8 + 6 + 10}, \frac{8 \times 6 + 6 \times 0 + 10 \times 0}{8 + 6 + 10} \right) \equiv (2, 2).$$

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Q7

(4)



Since, $x - y + 3 = 0$ and $x + y = 3$ are perpendicular lines and intersection point of $x - y + 3 = 0$ and $x + y = 3$ is $P(0, 3)$.

$$\Rightarrow M \text{ is mid-point of } PR \Rightarrow R(4, 5)$$

$$\text{Let } S(x_1, x_1 + 3) \text{ and } Q(x_2, 3 - x_2)$$

M is mid-point of SQ

$$\Rightarrow x_1 + x_2 = 4, x_1 + 3 + 3 - x_2 = 8$$

$$\Rightarrow x_1 = 3, x_2 = 1$$

Then, the vertex D is (3, 6).

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Q8

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(2) Since, $m_{QR} \times m_{PH} = -1$

$$\Rightarrow m_{QR} = -\frac{1}{m_{PH}}$$

$$\Rightarrow m_{QR} = \frac{y-3}{x-4} = 0$$

$$\Rightarrow y = 3$$

$$m_{PQ} \times m_{RH} = -1$$

$$\Rightarrow \frac{1}{4} \times \frac{y}{x} = -1$$

$$\Rightarrow y = -4x$$

$$\Rightarrow x = -\frac{3}{4}$$

$$\text{Vertex } R \text{ is } \left(-\frac{3}{4}, 3 \right)$$

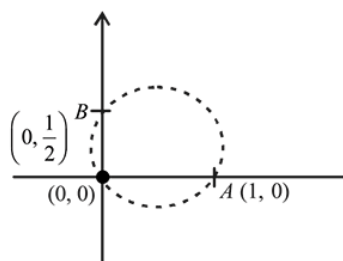
Hence, vertex R lies in second quadrant.

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Q9

Point and Straight Line

(1)



Let equation of circle be $x^2 + y^2 + 2gx + 2fy = 0$

As length of intercept on x axis is $1 = 2\sqrt{g^2 - c}$

$$\Rightarrow |g| = \frac{1}{2}$$

length of intercept on y -axis $= \frac{1}{2} = 2\sqrt{f^2 - c}$

$$\Rightarrow |f| = \frac{1}{4}$$

Equation of circle that passes through given points is

$$x^2 + y^2 - x - \frac{y}{2} = 0$$

Tangent at $(0, 0)$ is,

$$(y - 0) = \left(\frac{dy}{dx} \right)_{(0,0)} \cdot (x - 0)$$

$$\Rightarrow 2x + y = 0$$

Perpendicular distance from $B(1, 0)$ on the tangent to the

$$\text{circle} = \frac{1}{\sqrt{5}}$$

Perpendicular distance from $B\left(0, \frac{1}{2}\right)$ on the tangent to

$$\text{the circle} = \frac{2}{\sqrt{5}}$$

$$\text{Sum of perpendicular distance} = \frac{1}{2} + 2 = \frac{\sqrt{5}}{2}$$

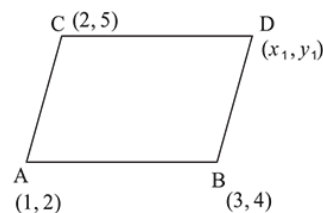
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Q10

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(1) Since, in parallelogram mid points of both diagonals coincide.

$\therefore$ mid-point of AD = mid-point of BC



$$\left(\frac{x_1+1}{2}, \frac{y_1+2}{2} \right) = \left(\frac{3+2}{2}, \frac{4+5}{2} \right)$$

$$\therefore (x_1, y_1) = (4, 7)$$

Then, equation of AD is,

$$y - 7 = \frac{2-7}{1-4} (x - 4)$$

$$y - 7 = \frac{5}{3} (x - 4)$$

$$3y - 21 = 5x - 20$$

$$5x - 3y + 1 = 0$$

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Q11

(2) Let $\vec{v}_1$ and $\vec{v}_2$ be the vectors perpendicular to the plane OPQ and PQR respectively.

$$\vec{v}_1 = \overrightarrow{PQ} \times \overrightarrow{OQ} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 2 & 1 & 3 \end{vmatrix}$$

$$= 5\hat{i} - \hat{j} - 3\hat{k}$$

$$\vec{v}_2 = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ -2 & -1 & 1 \end{vmatrix}$$

$$= \hat{i} - 5\hat{j} - 3\hat{k}$$

$$\therefore \cos \theta = \frac{\vec{v}_1 \cdot \vec{v}_2}{|\vec{v}_1| |\vec{v}_2|} = \frac{5+5+9}{25+1+9} = \frac{19}{35}$$

$$\therefore \theta = \cos^{-1} \left(\frac{19}{35} \right)$$

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Point and Straight Line

Q12

(4) $\therefore$ Equation of straight line can be rewritten as,

$$y = \frac{2}{3}x + \frac{17}{3}.$$

$\therefore$ Slope of straight line $= \frac{2}{3}$

Slope of line passing through the points (7, 17) and

$$(15, \beta) = \frac{\beta - 17}{15 - 7}$$

$$= \frac{\beta - 17}{8}$$

Since, lines are perpendicular to each other.

Hence, $m_1 m_2 = -1$

$$\Rightarrow \left(\frac{2}{3}\right)\left(\frac{\beta - 17}{8}\right) = -1$$

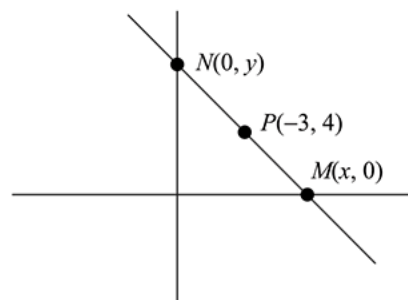
$$\Rightarrow \beta = 5$$

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Q13

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(2) Since, P is mid point of MN



$$\text{Then, } \frac{0 + x}{2} = -3$$

$$\Rightarrow x = -3 \times 2$$

$$\Rightarrow x = -6$$

$$\text{and } \frac{y + 0}{2} = 4 \Rightarrow y + 0 = 2 \times 4$$

$$\Rightarrow y = 8$$

Hence required equation of straight line MN is

$$\frac{x}{-6} + \frac{y}{8} = 1$$

$$\Rightarrow 4x - 3y + 24 = 0$$

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Q14

(3) A point which is equidistant from both the axes lies on either $y = x$ and $y = -x$.

Since, point lies on the line $3x + 5y = 15$

Then the required point

$$3x + 5y = 15$$

$$\frac{x + y = 0}{x = -\frac{15}{2}}$$

$$y = \frac{15}{2} \Rightarrow (x, y) = \left(-\frac{15}{2}, \frac{15}{2}\right) \{2^{\text{nd}} \text{ quadrant}\}$$

$$\text{or } 3x + 5y = 15$$

$$\frac{x - y = 0}{x = \frac{15}{8}}$$

$$y = \frac{15}{8} \Rightarrow (x, y) = \left(\frac{15}{8}, \frac{15}{8}\right) \{1^{\text{st}} \text{ quadrant}\}$$

Hence, the required point lies in 1st and 2nd quadrant.

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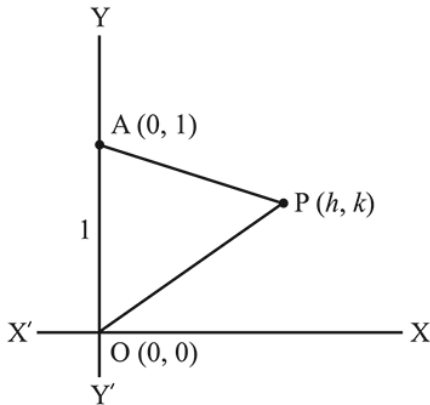
Q15

(3) Let point $P(h, k)$

$$\therefore OA = 1$$

$$\text{So, } OP + AP = 3$$

$$\Rightarrow \sqrt{h^2 + k^2} + \sqrt{h^2 + (k-1)^2} = 3$$



$$\Rightarrow h^2 + (k-1)^2 = 9 + h^2 + k^2 - 6\sqrt{h^2 + k^2}$$

$$\Rightarrow 6\sqrt{h^2 + k^2} = 2k + 8$$

$$\Rightarrow 9h^2 + 8k^2 - 8k - 16 = 0$$

Hence, locus of point P is

$$9x^2 + 8y^2 - 8y - 16 = 0$$

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Q16

(1) $\therefore (h, k), (1, 2)$ and $(-3, 4)$ are collinear

$$\therefore \begin{vmatrix} h & k & 1 \\ 1 & 2 & 1 \\ -3 & 4 & 1 \end{vmatrix} = 0 \Rightarrow -2h - 4k + 10 = 0$$

$$\Rightarrow h + 2k = 5 \quad \dots(i)$$

$$\text{Now, } m_{L_1} = \frac{4-2}{-3-1} = -\frac{1}{2} \Rightarrow m_{L_2} = 2 \quad [\because L_1 \perp L_2]$$

By the given points (h, k) and $(4, 3)$,

$$m_{L_2} = \frac{k-3}{h-4} \Rightarrow \frac{k-3}{h-4} = 2 \Rightarrow k-3 = 2h-8$$

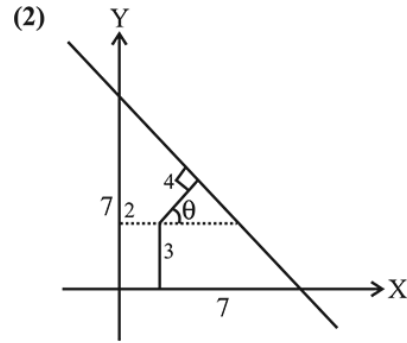
$$2h - k = 5 \quad \dots(ii)$$

From (i) and (ii)

$$h = 3, k = 1 \Rightarrow \frac{k}{h} = \frac{1}{3}$$

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Q17



Since point at 4 units from $P(2, 3)$ will be $A(4 \cos \theta + 2, 4 \sin(\theta + 3))$ and this point will satisfy the equation of line $x + y = 7$

$$\Rightarrow \cos \theta + \sin \theta = \frac{1}{2}$$

On squaring

$$\Rightarrow \sin 2\theta - \frac{3}{4} \Rightarrow \frac{2 \tan \theta}{1 + \tan^2 \theta} = -\frac{3}{4}$$

$$\Rightarrow 3 \tan^2 \theta + 8 \tan \theta + 3 = 0$$

$$\Rightarrow \tan \theta = \frac{-8 \pm 2\sqrt{7}}{6} \quad (\text{ignoring } -ve \text{ sign})$$

$$\Rightarrow \tan \theta = \frac{-8 + 2\sqrt{7}}{6} = \frac{1 - \sqrt{7}}{1 + \sqrt{7}}$$

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Q18

(1) $\therefore$ two lines are perpendicular $\Rightarrow m_1 m_2 = -1$

$$\Rightarrow \left(\frac{-1}{a-1} \right) \left(\frac{-2}{a^2} \right) = -1$$

$$\Rightarrow 2 = a^2(1-a) \Rightarrow a^3 - a^2 + 2 = 0$$

$$\Rightarrow (a+1)(a^2 + 2a + 2) = 0 \Rightarrow a = -1$$

Hence equations of lines are $x - 2y = 1$ and $2x + y = 1$

$$\therefore \text{intersection point is } \left(\frac{3}{5}, \frac{-1}{5} \right)$$

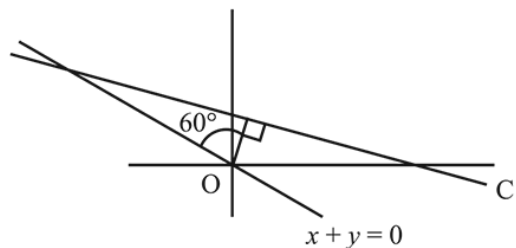
$$\text{Now, distance from origin} = \sqrt{\frac{9}{25} + \frac{1}{25}} = \sqrt{\frac{10}{25}} = \sqrt{\frac{2}{5}}$$

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Q19

(2, 4) $\therefore$ perpendicular makes an angle of 60° with the line $x + y = 0$.

$\therefore$ the perpendicular makes an angle of 15° or 75° with x -axis.



Hence, the equation of line will be

$$x \cos 75^\circ + y \sin 75^\circ = 4$$

$$\text{or } x \cos 15^\circ + y \sin 15^\circ = 4$$

$$(\sqrt{3} - 1)x + (\sqrt{3} + 1)y = 8\sqrt{2}$$

$$\text{or } (\sqrt{3} + 1)x + (\sqrt{3} - 1)y = 8\sqrt{2}$$

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